9.3 Similar Right Triangles

Name _____ Period ____ Date _____

Geometry 9.3 (30-minute plan) · Student sheet

I can…

- show that the altitude to the hypotenuse makes three similar right triangles.
- find the geometric mean of two numbers.
- use the Geometric Mean Theorems to find missing lengths.

Vocabulary · fill in the missing word

Altitude to the hypotenuse — the segment from the right angle _____ to the hypotenuse

Geometric mean — of two _____ numbers a and b: the positive number x with $x^2 = ab$, so $x = \sqrt{ab}$

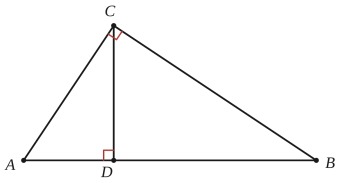
Similar triangles — triangles whose _____ angles are congruent and whose corresponding sides are proportional

Part 1 · Notes and examples (fill in each note, then work the example)

Right Triangle Similarity Theorem: the altitude to the hypotenuse of a right triangle forms two triangles that are _____ to the original triangle and to each other.

Example 1

In right △ABC, $\overline{CD}$ is the altitude to the hypotenuse. Name the three similar triangles.

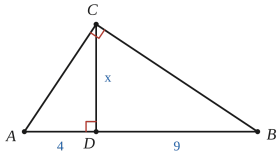


Geometric mean of a and b: x = _____

Geometric Mean (Altitude) Theorem: the altitude to the hypotenuse is the geometric mean of the two _____ of the hypotenuse: $\text{altitude}^2 = (\text{one segment})(\text{other segment})$.

Example 2

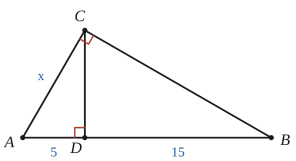
The altitude to the hypotenuse divides it into segments of 4 and 9. Find the length of the altitude, x.



Geometric Mean (Leg) Theorem: each leg is the geometric mean of the _____ and the segment of the hypotenuse _____ that leg: $\text{leg}^2 = (\text{hypotenuse})(\text{segment next to the leg})$.

Example 3

The altitude to the hypotenuse divides it into segments of 5 and 15. Find the length of the leg next to the segment of 5, x.



Part 2 · Partner practice

Partner practice

The altitude to the hypotenuse divides it into segments of 2 and 8. Find the altitude, and the leg next to the segment of 8.

